



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



Representations (T2MP)

Index

- **Theory Recap**
- Informal and Formal languages
- Representing entities
- Representing knowledge
- Representing concepts
- Dtypes and Etypes

Theory Recap - Mental activities

- **Perception.** From **reality** to, e.g., the **images** of a “smiling face”, the “face of an old/young woman”, suitably composed, e.g., the “**complex**” **images** of “a smiling face in a plug”, “an old woman looking down and left to something”.
- **Conceptualization.** From, e.g., **images** to **words**, that is, an **alphabet**, ... **naming** what we perceive (e.g., the words: “left to”, “bumped”, “face”, “woman”, “young”).
- **Understanding.** From **words** to **sentences** describing the complex combinations, e.g., complex images, that we perceive, e.g., the sentences: “the plug looks like a smiling face”, “people live”, “an old woman at the bottom and a young woman at the bottom”.
- **Reasoning:** From **sentences** that describe what we have perceived to **new sentences** which describe what **must** be the case as a necessary **consequence** of what we have perceived **because learned in time**, and so on, recursively.

Theory Recap - Semantic Gap

Intuition (Semantic Gap). The **Semantic Gap** is the phenomenon which happens because of the impossibility for humans and machines to perceive the world in the same way (as it really is?).

Observation (misalignment of representations). The Semantic Gap is the source of the **pervasive misalignment** of the mental models of the world that **humans**, and also **machines**, build.

Intuition (Semantic gap) The **semantic gap** is the difference between the world and a human's mental representation of the world itself, what (s)he has perceived, as witnesses by the multiple different personal mental representations.

Theory Recap - Mental Representations

Intuition (Analogical mental representations) Analogical mental **representations** are mental representations that **depict** the world as we perceive it through *perception*. They are complex articulations of **percepts** into **facts**.

Intuition (Linguistic mental representations) Linguistic mental **representations** are mental representations that **describe** mental analogical representations using language. They use **assertions** formed using the alphabet and formation rules of a selected assertional **language**.

Theory Recap - Representations

Intuition 2.10 (Representations) A **representation** is a part of the world, developed by a human, that represents that human's mental representation of the world

Representations are **accessible**, via one of the five senses, to other humans.

A representation can be perceived, in the same way as the reality it represents. The perception of a representation and of its represented reality can be compared for similarity checking.

Intuition 2.11 (Analogical Representations) Analogical representations **depict** analogical mental representations.

Intuition 2.12 (Linguistic Representations) Linguistic representations **describe** analogic mental representations.

Theory Recap - Use of Representations

Observation (Requirement on representations). Representations are built with the goal of minimizing the probability of different interpretations and, therefore, of mental representations.

Observation (Using representations). Different interpretations may still arise. Risk minimized (not eliminated) via software and knowledge engineering methodologies.

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Formal Grammar

A **Formal Grammar** is a set of symbols and production rules used to perform substitutions on the given symbols to generate strings of a **Formal Language**

Example: Language of all binary numbers

Non-terminal Symbols : $\{A, B\}$

Terminal Symbols : $\{1, 0\}$

Starting Symbol : A

Production rules:

$A \rightarrow 1A \mid 1B \mid 1$

$B \rightarrow 0B \mid 0A \mid 0$

Informal and Formal languages

An **Informal Language** is any kind of language that doesn't have a formal grammar.

Examples of informal languages include English, Italian, and any other natural language.

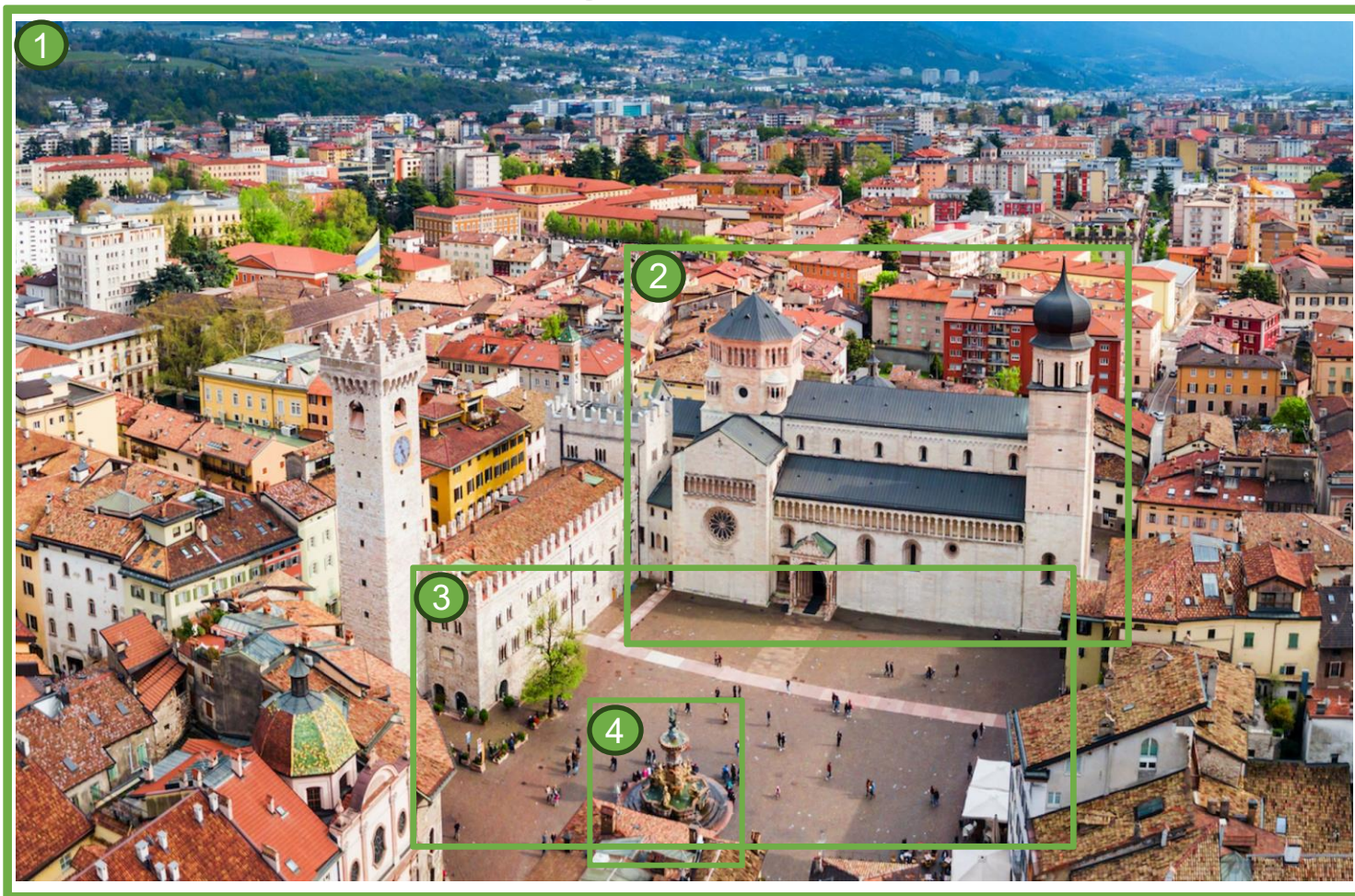
A **Formal Language** is any language that has a formal grammar.

Examples of formal languages are: Regex, any Programming Language, and UML.

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Representing entities



Etypes: City, Church, Square, Fountain

Entities: Trento, Duomo, Piazza Duomo, Fontana del Nettuno

Relations: Is in

Dtypes: number, bool

Values: 158000, 120000, 2200, 5760, false

Properties: area, number people, potable

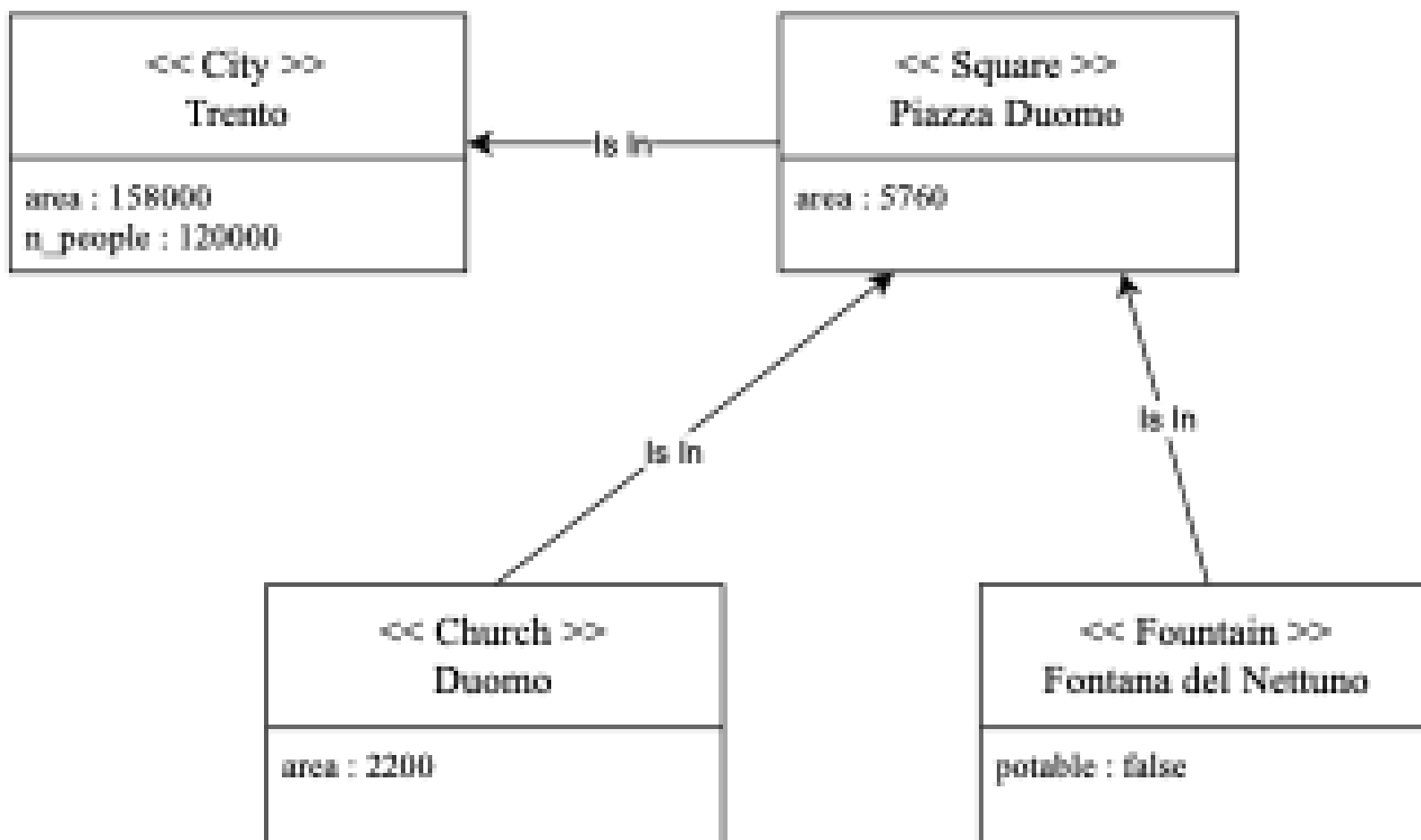
Representing entities - Tables

City		
name	area	n_people
Trento	158000	120000

Fountain		
name	potable	is in
Fontana del Nettuno	false	Piazza Duomo

Locations			
name	kind	is in	area
Duomo	Church	Piazza Duomo	2200
Piazza Duomo	Square	Trento	5760

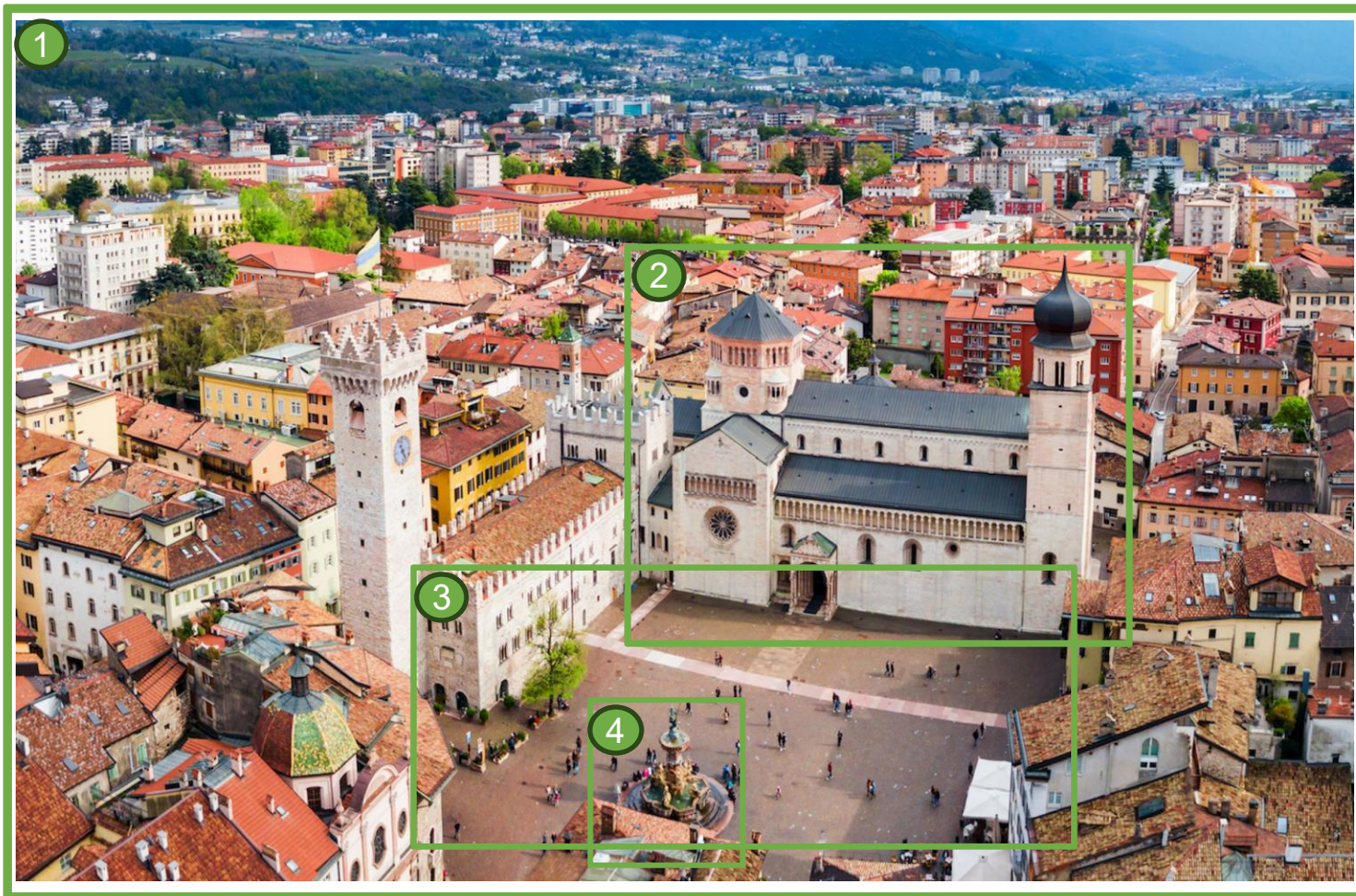
Representing entities - Graph



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Representing knowledge



Etypes: City, Church, Square, Fountain

Entities: ~~Trento, Duomo, Piazza Duomo, Fontana del Nettuno~~

Relations: Is in

Dtypes: number, bool

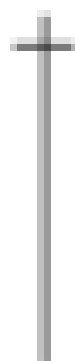
Values: ~~158000, 120000, 2200, 5760, false~~

Properties: area, number people, potable

ER Notation

Entity-Relationship diagram with IDEF1X and Crow's foot notation

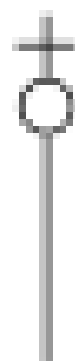
etype	
PK	<u>identifier : type</u>
	property1 : type
	property2 : type



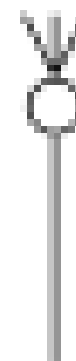
1



N



0-1



0-N

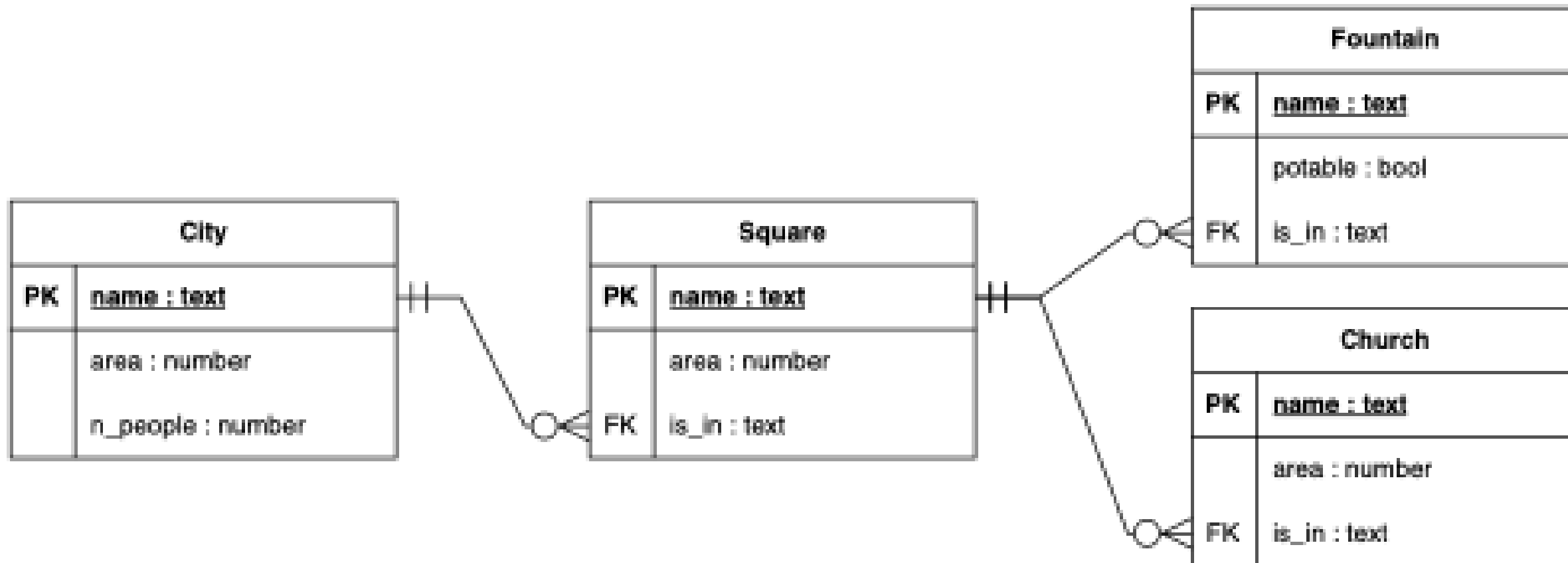


only 1

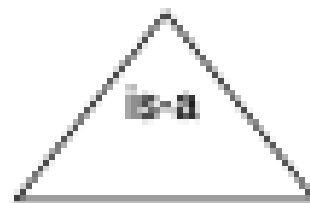


1-N

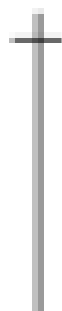
Representing knowledge - ER



EER Notation



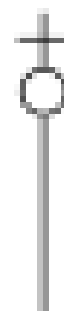
etype	
PK	<u>identifier : type</u>
	property1 : type
	property2 : type



1



N



0-1



0-N

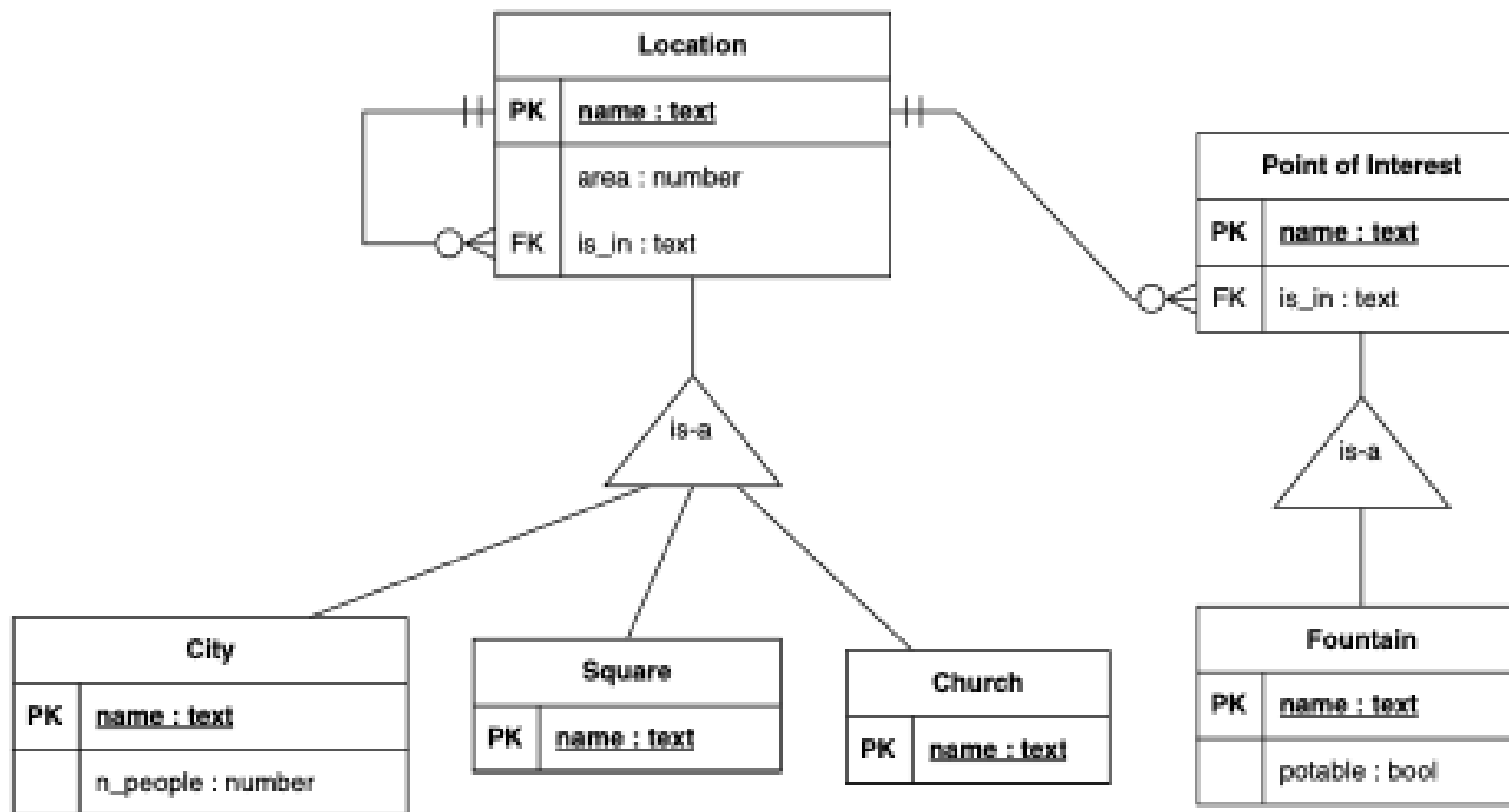


only 1



1-N

Representing knowledge - EER



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Representing concepts - WordNet

Wordnet a **digitalized** lexicon which encodes how the meaning of words (concepts) is connected, so-called **Lexical-Semantics**. **Built to be used by machines**.
It organizes each word in sets of words with closely related meanings (**synsets**).

It captures the semantic relationship between synsets:

- **Hypernyms**: Broader, more general concept.
- **Hyponyms**: Instance or subtype of a concept.
- **Antonyms**: Two words with contrasting meaning.
- **Meronyms**: Denote components or parts of a whole.
- **Holonyms**: Refers to the whole itself.

Representing concepts - «City»

Sense 1

- city**, metropolis, urban center -- (a large and densely populated urban area; may include several independent administrative districts; "Ancient Troy was a great city")
- => municipality -- (an urban district having corporate status and powers of self-government)
 - => urban area, populated area -- (a geographical area constituting a city or town)
 - => geographical area, geographic area, geographical region, geographic region -- (a demarcated area of the Earth)
 - => region -- (a large indefinite location on the surface of the Earth; "penguins inhabit the polar regions")
 - => location -- (a point or extent in space)
 - => object, physical object -- (a tangible and visible entity; an entity that can cast a shadow; "it was full of rackets, balls and other objects")
 - => physical entity -- (an entity that has physical existence)
 - => entity -- (that which is perceived or known or inferred to have its own distinct existence (living or nonliving))
 - => administrative district, administrative division, territorial division -- (a district defined for administrative purposes)
 - => district, territory, territorial dominion, dominion -- (a region marked off for administrative or other purposes)
 - => region -- (a large indefinite location on the surface of the Earth; "penguins inhabit the polar regions")
 - => location -- (a point or extent in space)
 - => object, physical object -- (a tangible and visible entity; an entity that can cast a shadow; "it was full of rackets, balls and other objects")
 - => physical entity -- (an entity that has physical existence)
 - => entity -- (that which is perceived or known or inferred to have its own distinct existence (living or nonliving))

Representing concepts - «Church»

Sense 2

church, church building -- (a place for public (especially Christian) worship; "the church was empty")

=> place of worship, house of prayer, house of God, house of worship -- (any building where congregations gather for prayer)

=> building, edifice -- (a structure that has a roof and walls and stands more or less permanently in one place; "there was a three-

=> structure, construction -- (a thing constructed; a complex entity constructed of many parts; "the structure consisted of a series of ribbons")

=> artifact, artefact -- (a man-made object taken as a whole)

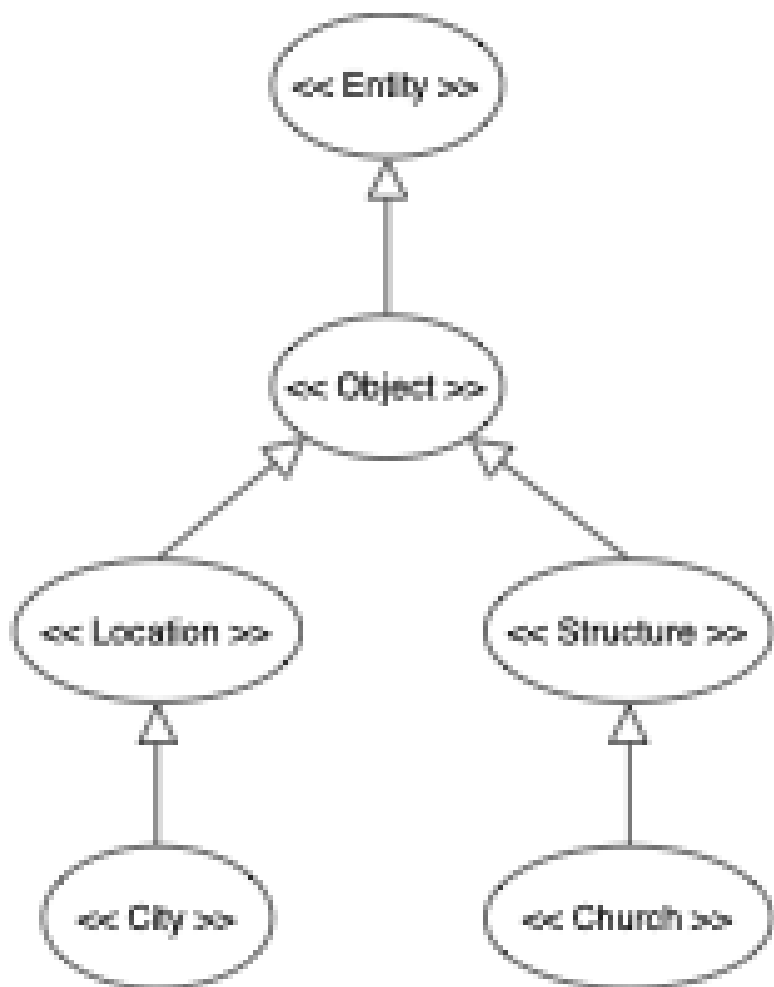
=> whole, unit -- (an assemblage of parts that is regarded as a single entity; "how big is that part compared to the whole?")

=> object, physical object -- (a tangible and visible entity; an entity that can cast a shadow; "it was full of rackets, balls and bats")

=> physical entity -- (an entity that has physical existence)

=> entity -- (that which is perceived or known or inferred to have its own distinct existence (living or nonliving))

Representing concepts



Observation. We can represent a concept's Hyponyms hierarchy as a tree with << Entity >> as its root. In the representation, we can keep only the nodes that are important for our purpose.

Observation. Note how << Church >> is not a hyponym of "Location" in WordNet. In this case either the lexicon has to be extended to support the use case, or the knowledge layer has to be adjusted to follow the concept layer.

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Etypes

Definition (Etype). An Etype is a set of multiple Entities that look the same.

- Two Entities of the same Etype have a subset of their properties in common (not necessarily with the same values).
- Two Entities of the same Etype can be in the same types of relation.
- An Etype can be a subset of another Etype, inheriting its properties.
- **Entity** is the most general Etype, which contains all entities; any other Etype is a special case of the Etype Entity.

Example (Etype). Some examples of Etypes are: Person, Building, City, Square, Teacher, Student, Parent, Son ...

Dtypes

Definition (Datatype, value). A data type is an etype where data and object properties are defined a priori. A data type is defined by providing:

- (i) the names of its values;
- (ii) data properties and their values;
- (iii) object properties and their values;
- (iv) the notion of equality between values

Example (Natural Number). We have the following

- Values: numerals, written 1,2,3, ..., denoting natural numbers, 1,2,3,...
- Data properties: odd, even, prime, multiple
- Object properties: <, >, plus, minus, times, division
- Equality: =

Example (Data type). Number, real, string, identifier, date, time, coordinates, weight, data (this being the most general datatype).

Observation (Data type, data value). Data types must be used “as is”. The use of data types is a well established practice in Computer Science.



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